

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

8875/02

Paper 2 Core Paper

15 September 2014

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	11
2	21
3	8
Section B	
	20
Total	60

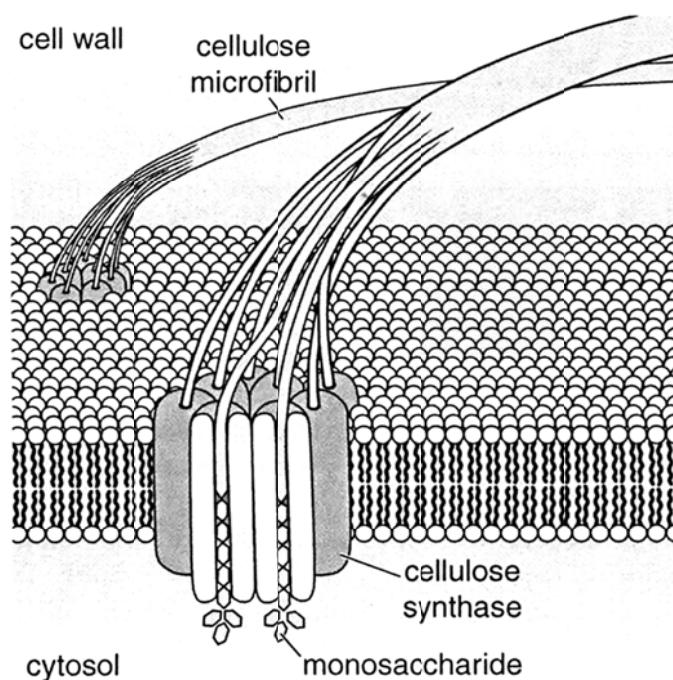
This document consists of **10** printed pages.



Section AAnswer **all** questions.

- 1** Plant cells have a cellulose cell wall outside the cell surface membrane. The cellulose microfibrils are embedded in a matrix of hemicellulose, pectin and proteins.

The cell surface membrane of a plant cell contains the enzyme, cellulose synthase, which synthesises microfibrils. It catalyses bond formation between monosaccharides in the cytosol. The monosaccharides pass through the enzymes and leave the cell as they are joined to form microfibrils. This is shown in Fig. 1.

**Fig. 1**

- (a)** **(i)** Name the monosaccharide units that linked together to form cellulose.
 [1]
- (ii)** Name the bond between monosaccharide in cellulose catalysed by cellulose synthase.
 [1]
- (iii)** Describe how the bond in **(ii)** is formed. You may use an annotated diagram in the space below.

[2]

- (b)** Explain how cellulose synthase speeds up the formation of microfibrils.

.....

.....

.....

.....

.....

..... [3]

- (c)** Explain why cellulose cannot be synthesised inside the cell.

.....

..... [1]

Cellulose fibres are made up of many cellulose microfibrils. Each microfibril is made up of many cellulose molecules lying parallel to each other.

- (d)** With reference to Fig. 1 and your knowledge of the structure of cellulose, explain how cellulose is suited to perform its function.

.....

.....

.....

.....

.....

..... [3]

[Total:11]

- 2 Fig. 2.1 shows ribosomes attached to the endoplasmic reticulum, carrying out insulin synthesis.

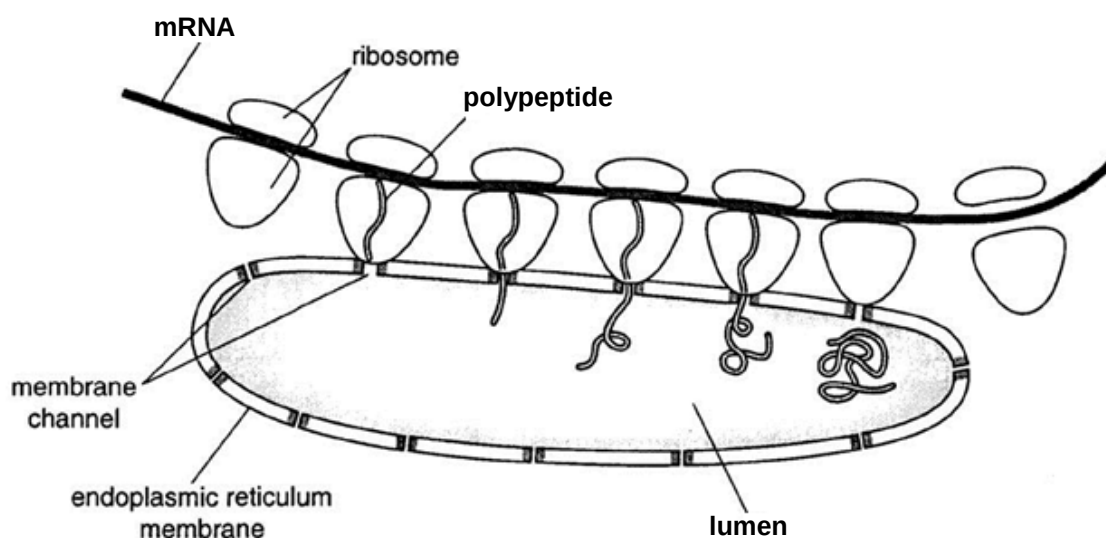


Fig. 2.1

- (a) With reference to Fig. 2.1, describe the function of the membrane channel.

.....

.....

.....

..... [2]

- (b) The polypeptide entering the lumen is processed before it leaves the endoplasmic reticulum. Describe the post-translational processing of the polypeptide to form insulin.

.....

.....

.....

..... [2]

(c) Insulin synthesised is secreted into the bloodstream. Presence of insulin in the bloodstream lowers blood glucose level. Thus production of insulin is highly regulated.

(i) Briefly describe the events leading up to the secretion of insulin after its processing in the lumen.

.....

.....

.....

.....

.....

..... [3]

(ii) Outline how human insulin can be produced by genetic engineering techniques.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

Fig. 2.2 shows the sequence of bases in a section of the mRNA in Fig. 2.1, coding for the first few amino acids of insulin.

3' AUGUUGCCCCCUAGGUA AU 5'

Fig. 2.2

Each codon codes for a specific amino acid as illustrated in Table 2,

Table 2

UUU Phe	UCU Ser	UAU Tyr	UGU Cys
UUC Phe	UCC Ser	UAC Tyr	UGC Cys
UUA Leu	UCA Ser	UAA End	UGA End
UUG Leu	UCG Ser	UAG End	UGG Trp
CUU Leu	CCU Pro	CAU His	CGU Arg
CUC Leu	CCC Pro	CAC His	CGC Arg
CUA Leu	CCA Pro	CAA Gln	CGA Arg
CUG Leu	CCG Pro	CAG Gln	CGG Arg
AUU Ile	ACU Thr	AAU Asn	AGU Ser
AUC Ile	ACC Thr	AAC Asn	AGC Ser
AUA Ile	ACA Thr	AAA Lys	AGA Arg
AUG Met	ACG Thr	AAG Lys	AGG Arg
GUU Val	GCU Ala	GAU Asp	GGU Gly
GUC Val	GCC Ala	GAC Asp	GGC Gly
GUA Val	GCA Ala	GAA Glu	GGA Gly
GUG Val	GCG Ala	GAG Glu	GGG Gly

- (d) (i) Using information from Table 2, list the corresponding amino acids coded by the mRNA in Fig. 2.2.

..... [2]

- (ii) Explain why each codon consists of three bases.

.....

.....

.....

.....

..... [2]

A ten amino acids long section of the polypeptide chain was partially hydrolysed to release oligopeptides shown in Fig. 2.2.

gly-ala-gly-val

gly-gly-lys

gly-val-ile-phe

phe-thr-gly-gly

Fig. 2.3

- (e) (i) With reference to Fig. 2.3, work out the order of amino acids in the polypeptide.

..... [1]

- (ii) In the space provided, illustrate the hydrolysis of a peptide bond between two glycine residues.

[2]

- (f) (i) Explain why single base deletions in a coding DNA sequence are more likely to produce non-functional proteins than single base substitutions.

..... [2]

- (ii) Explain why some base substitutions do not lead to change in the protein coded for.

..... [1]

[Total: 21]

3 The inheritance of a genetic skin disorder in a family is shown in Fig. 3.

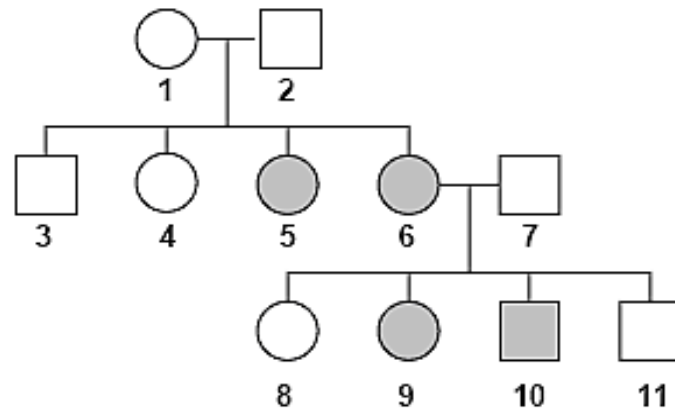


Fig. 3

(a) With reference to Fig. 3, explain the mode of inheritance of this disease.

.....

.....

.....

..... [2]

(b) Blood groups of several individuals are listed below.

- Individual 1 – blood group O
- Individual 5 – blood group B
- Individual 6 – blood group A
- Individual 7 – blood group AB

(i) State the blood group of individual 2.

..... [1]

(ii) State the probability that individual 4 has blood group A.

..... [1]

- (c) Using appropriate symbols, illustrate the probability that individual **6** and **7** will have a child with blood group A who suffers from the skin disorder.

[4]

[Total: 8]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

- 4 **(a)** Explain, with appropriate examples, how natural selection brings about evolution. [7]
- (b)** Explain how homology supports Darwin's theory of natural selection. [7]
- (c)** Describe the sources of genetic variation in a population. [6]
- [Total: 20]
- 5 **(a)** Describe how the Calvin cycle differs from the Krebs cycle. [6]
- (b)** Explain the small yield of ATP under anaerobic conditions in both yeast and mammals. [8]
- (c)** Explain how the structure of mitochondria and chloroplasts supports the theory that these organelles are of prokaryotic origin. [6]

[Total: 20]